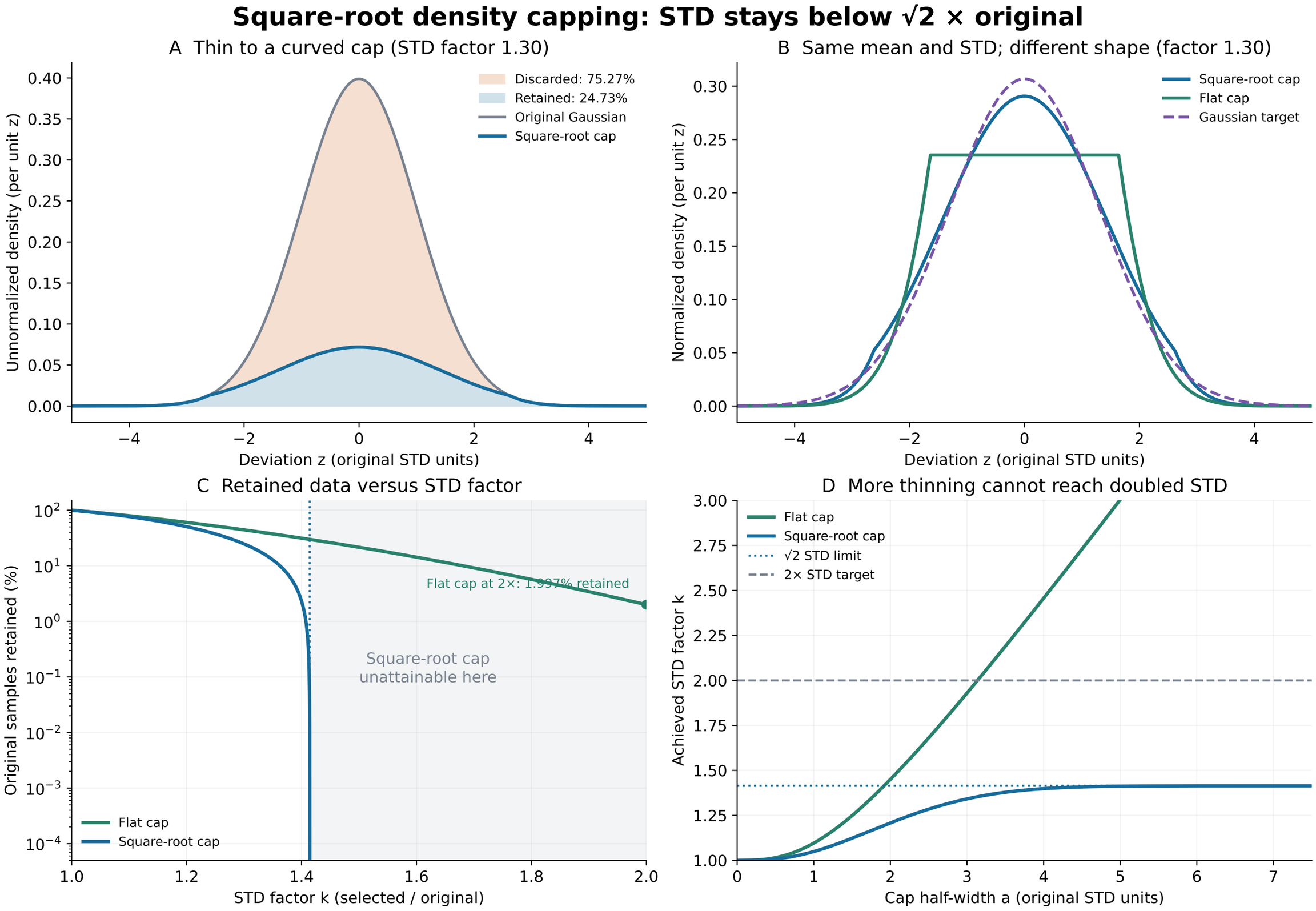

Square-root density capping: STD stays below √2 × original
A Thin to a curved cap (STD factor 1.30)
Discarded: 75.27%
Retained: 24.73%
Original Gaussian
Square-root cap
Unnormalized density (per unit z)
Deviation z (original STD units)
B Same mean and STD; different shape (factor 1.30)
Square-root cap
Flat cap
Gaussian target
Normalized density (per unit z)
Deviation z (original STD units)
C Retained data versus STD factor
Flat cap at 2×: 1.997% retained
Square-root cap
unattainable here
Flat cap
Square-root cap
Original samples retained (%)
STD factor k (selected / original)
D More thinning cannot reach doubled STD
Flat cap
Square-root cap
√2 STD limit
2× STD target
Achieved STD factor k
Cap half-width a (original STD units)